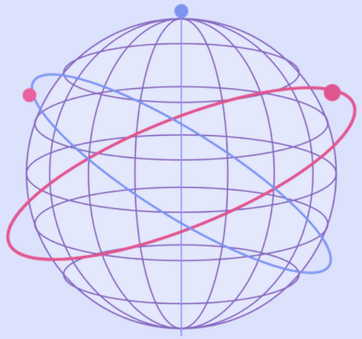
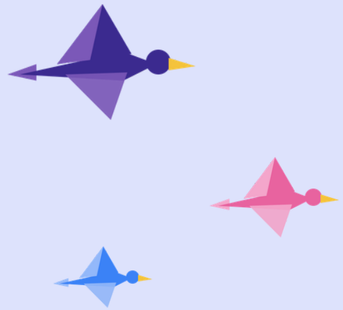
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
QUANTA — Quantum-Assisted Timetabling & Allocation

TRACK
Both (Classical + Quantum)

TEAM NAME
Team VTU-QFF-HC-0076

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INSTITUTION / ORGANISATION
Visvesvarayya technological University Belagavi

COUNTRY
India

Problem Statement

THE REAL-WORLD PROBLEM

Imagine you are the chief examination coordinator at a university with 4,200 students, 180 courses, 22 examination halls and 15 time slots across one exam week.

Hard constraints — zero tolerance

No clashes for any student
Seats ≥ students in every exam
No hall double-booking
Every course has a slot and hall

Soft constraints — wellbeing

Back-to-back exams: fatigue
3 exams in a day: overload
Uneven spread across the week

Why existing methods fail

Classical alone: tangled electives blow up runtime
Quantum alone: 100–156 qubits; campus needs tens of thousands

Objectives & Novelty

Objectives

1. **Zero hard violations**
   Clash-free, capacity-safe, fully covered timetable
2. **Cut student fatigue**
   Lower back-to-back and 3+ exams-a-day penalties
3. **Fit NISQ limits**
   Quantum only on the worst ≤ 12-course bottleneck
4. **Prove it honestly**
   Benchmark against CP-SAT, brute force, annealing

Novelty

1. **Quantum proposes, classical disposes**
   Deterministic acceptance gate has the last word
2. **1 qubit per course**
   Perturbation encoding: stay in slot A or move to slot B
3. **Independent auditor**
   Pure-Python scorer — no OR-Tools or Qiskit imports
4. **One pipeline, three backends**
   Ideal sim, noisy Aer, IBM Kingston QPU

Solution & Quantum Advantage

The Hybrid Solution (How It Works)

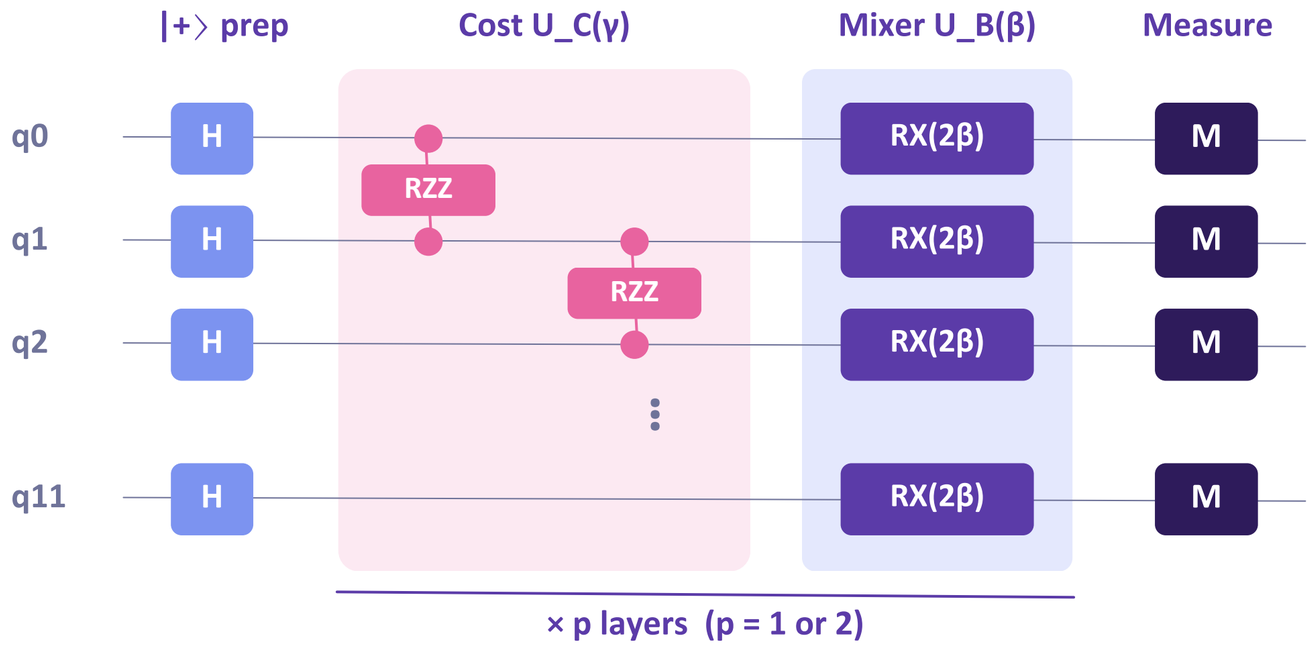
1. **Classical Baseline**
   Solves the full campus schedule (4,200 students, 180 courses) with **0 exam clashes**.
2. **Bottleneck Isolation**
   Extracts the single toughest conflict cluster (10–12 courses causing back-to-back exam fatigue).
3. **Quantum Optimization (QAOA)**
   Explores slot combinations in superposition to reduce student stress.
4. **Classical Safety Gate**
   Verifies the quantum result; accepts it **only** if it improves the schedule with 0 clashes, else falls back to baseline.

The Quantum Advantage (Why It Matters)

5,000+ → 12 binary variables → qubits

- **Massive qubit compression**
  Shrinks a 5,000+ variable problem into just 12 qubits (1 qubit = 1 course move), fitting today's IBM hardware.
- **Untangles soft conflicts**
  Targets dense knots of back-to-back exams where classical solvers often stall.
- **100% error-proof output**
  Hardware noise can't reach students: the gate guarantees a mathematically clash-free schedule.

Algorithm & Circuit Design



COBYLA / SPSA updates (γ, β) to minimise measured energy

Key details

QUANTA Algorithm

1. Classical Baseline (CP-SAT):Solves full campus schedule with 0 clashes.

2. Isolate Bottleneck (NetworkX):Extracts worst conflict cluster (**≤ 12 courses).

3. 1-Qubit Mapping (QUBO): Assigns 1 qubit per course (`0` = Stay in Slot A, `1` = Move to Slot B).

4. Quantum Search (QAOA):Explores slot combinations in superposition to minimize student fatigue.

5. Classical Repair: Matches rooms (Hopcroft-Karp) and rolls back invalid moves.

6. Safety Gate: Deploys only if Clashes = 0& fatigue improves; else keeps baseline.

Implementation & Tech Stack

1 Data Ingestion & Baseline

Parses 4 CSVs, solves the 4,200-student schedule with **0 hard violations** (CP-SAT).

2 Bottleneck Extraction

NetworkX conflict graph isolates the ≤ 12-course friction cluster and a candidate **Slot B**.

3 Quantum Execution

12-qubit Ising HC + **QAOA** (COBYLA / SPSA), 512–4,096 shots → **top 32** bitstrings.

4 Repair & Safety Gate

Hopcroft-Karp + rollback fix rooms; **Q-OPT** deploys only if hard = 0 and soft drops.

TECH STACK · LAYER	TECHNOLOGIES & TOOLS	CORE RESPONSIBILITY
Quantum Engine backend/quantum/	**Qiskit** · **Aer** · **Runtime** (SamplerV2, Estimator) · **IBM Kingston QPU**, 156-qubit Heron r2	QUBO→Ising, QAOA transpile (level 3, dynamical decoupling), T_1/T_2 noise sim, live QPU
Classical Engine backend/core/	**Python** · **Google OR-Tools (CP-SAT)** · **NetworkX** · **Hopcroft-Karp**	Global solve, conflict-graph clustering, room matching, zero-trust constraint audit
Backend & API backend/api/	**FastAPI** · **Pydantic v2** · **SQLite** · **Uvicorn** · **Pytest** (61 tests)	REST endpoints, schema validation, async job queue for QPU runs, telemetry caching
Frontend Console frontend/	**React** · **TypeScript** · **Vite** · **D3.js** · **Recharts** · **Bento Grid UI**	Master timetable, before/after conflict graph, live energy traces, hardware telemetry

Results & Analysis

0

hard violations reach the final schedule

< 40 ms

classical exact solve at N ≤ 12

Seconds

QAOA runtime on the same problem

Methods benchmarked · 10 random seeds

brute_force	cpsat	sim_anneal
qaoa_ideal	qaoa_noisy	qaoa_ibm

+ circuit scaling 6→12 qubits · IBM Kingston depth, 2-qubit gates, queue time

Analysis

- Classical exact methods win on speed at this size — we report it openly.
- Noisy or invalid quantum outputs are repaired or rejected; fallback keeps hard violations at 0.
- Accepted improvements are tagged Q-OPT in the master timetable.
- Value shown: safe integration, not raw speed-up.

QUANTA: Impact & Real-World Applications

1 Direct Campus Impact *University Timetabling*

- **Scales to real universities:** Effortlessly handles massive campus loads, tested on **4,200 students, 180 courses, 22 halls and 15 time slots**.
- **Boosts student wellbeing:** Actively targets and reduces severe mental fatigue by eliminating **back-to-back exams** and single-day exam overloads (3+ exams/day).
- **Solves cross-department knots:** Untangles dense conflict clusters caused by students taking interdisciplinary electives (e.g., CS students taking Bio-Ethics or Economics) where classical solvers stall.
- **100% zero-risk deployment:** The *Deterministic Acceptance Gate* mathematically guarantees **0 hard clashes**, zero double-booked halls and no room capacity overflows, even if quantum hardware crashes or produces noise.

2 Architectural Impact *NISQ-to-Production Blueprint*

- **Bridges the hardware gap:** Shrinks a 5,000+ variable scheduling problem down to **≤ 12 qubits** using 1-qubit-per-course perturbation encoding, making today's ~156-qubit processors (like IBM Heron) immediately useful.
- **Safe enterprise integration:** Proves how to deploy noisy quantum algorithms safely into mission-critical production workflows via a *"Quantum Proposes, Classical Disposes"* architecture.

3 Broader Real-World Applications

Where QUANTA's hybrid bottleneck-extraction engine can be applied next

INDUSTRY SECTOR	SCHEDULING BOTTLENECK	QUANTA HYBRID APPLICATION
Healthcare & Hospitals	Nurse shift fatigue & Operating Room (OR) clashes	Classical solver sets baseline hospital shifts; Quantum untangles peak-hour surgery bottlenecks and back-to-back ICU shifts.
Aviation & Transit	Crew burnout & gate/runway congestion	Classical baseline routes global fleets; Quantum resolves dense hub delays and minimizes tight crew layover turnarounds.
Cloud & Data Centers	Server overload & high-priority job collisions	Classical scheduler allocates standard workloads; Quantum optimizes high-friction GPU cluster bottlenecks during peak demand.
Manufacturing & Supply Chain	Assembly line bottlenecks & machine downtime	Classical engine plans factory floor schedules; Quantum re-sequences tight multi-part dependencies to prevent line stalls.

QUANTA: Challenges & Future Scope

Current Challenges & NISQ Limitatior

Qubit capacity bottleneck
Full campus needs 5,000+ binary variables—far beyond today's 100–156-qubit processors.

Hardware noise & decoherence
T_1 = 120 µs, T_2 = 90 µs, 2-qubit CNOT errors ≈ 10^{-2}, and 2% readout bit-flips.

Runtime latency vs. classical
For N ≤ 12, CP-SAT and brute force solve in <40 ms; QAOA takes seconds plus QPU queue time.

Single-hop move restriction
Each course can currently stay in baseline Slot A or move to one alternate Candidate Slot B.

Future Scope & Roadmap

Scale cluster sizes
Expand beyond ≤12 courses toward 50–100+ qubits as hardware fidelity and gate error rates improve.

Multi-slot quantum encoding
Move beyond Slot A/B binary perturbations to qudit or multi-qubit encodings for 4+ candidate slots.

Error mitigation & deeper QAOA
Integrate ZNE and PEC through Qiskit Runtime to support deeper circuits (p > 2) despite noise.

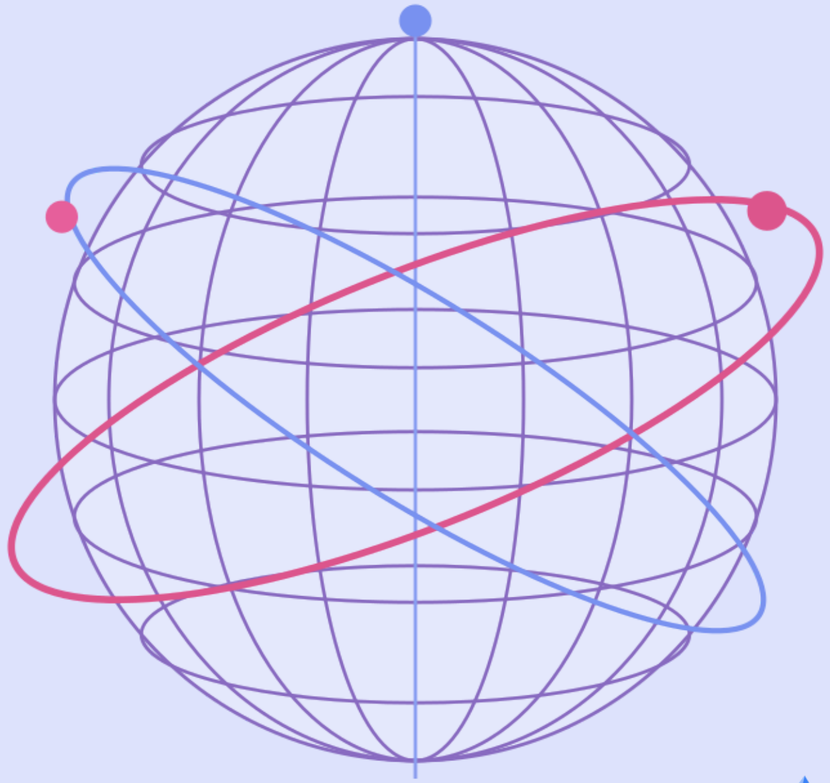
Parallel clusters & enterprise APIs
Optimize disjoint clusters concurrently; generalize the engine to hospital shifts, airline crews, and cloud GPU allocation.

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.



GITHUB REPOSITORY
github.com/your-team

DEMO VIDEO LINK
Paste link